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Variance Analysis and Flexible Budgeting

Managers often compare actual operating performance in a period to the expected or targeted performance. Expectations for performance can arise from several sources:

1. a budgeted amount from the financial plan
2. a standard, say for labor or materials usage, established by industrial engineers
3. last period's performance.

Accountants refer to the difference between actual and expected results as a **variance**.¹

Variance analysis enables managers to go beneath the surface of a reported difference between actual and expected results. While seldom producing definitive answers or explanations for such differences, variance analysis enables managers to focus and sharpen their questions that can then be answered with further investigation.

The basic idea behind variance analysis is extremely simple. An aggregate financial amount, such as a revenue or a cost, often can be decomposed into a quantity and a price component.

$$\text{Revenue or Cost} = \text{Price} * \text{Quantity} \quad (\text{symbolically, } R = P * Q; \text{ or } C = P * Q)$$

For example, expected revenue of \$60,000 from sale of a product can be decomposed into the number of units expected to be sold, 20,000, and the \$3 average price per unit:

$$\$60,000 = 20,000 * \$3$$

Electricity cost of \$18,000 in a period can be decomposed into the number of kilowatt hours expected to be used, 4,500, and the average cost per unit, \$4.00.

Variance analysis starts once actual results for a period are reported. For example, suppose actual revenue in a period was \$68,200, when budgeted revenues were \$60,000. The simplest variance analysis calculates:

¹ The accountant's variance, is similar in spirit (but completely different in calculation) to the statistician's variance. Both use the term to describe a measure of dispersion between actual and expected quantities.

Professor Robert S. Kaplan prepared this note as the basis for class discussion.

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$$\begin{aligned}
 \text{Total revenue variance} &= \text{Actual revenues} - \text{Budgeted revenues} \\
 &= \$68,200 - \$60,000 \\
 &= \$8,200 \text{ (favorable)}
 \end{aligned}$$

A variance is **favorable (F)** when it increases operating income and **unfavorable (U)** when it decreases operating income. Favorable variances occur when actual revenues exceed budgeted revenues, and when actual expenses are lower than expected. Revenues below expectations and expenses higher than expected lead to unfavorable variances.

Quantity and Price Variances

One hardly needs to take an accounting course at an expensive university to learn that when actual revenues exceed budgeted revenues, management will view this circumstance as a favorable event. To go beneath the simple analysis above, we need to know more about how the revenues were generated. Continuing our example, suppose that actual sales were 22,000 units and the average selling price was \$3.10 [verify that this combination will produce revenues of \$68,200]. How much of the \$8,200 is due to selling a higher quantity than was budgeted, and how much is due to the increase in average selling price? The **quantity variance** represents the contribution from actual unit sales differing from expected quantities. The difference is valued at the expected (budgeted) price. The **price variance** represents the difference caused by selling units at a price different from that expected or budgeted.

Symbolically (equations are more succinct and clearer than words in expressing these ideas), let the symbol A represent an actual quantity, and the symbol E represent an expected quantity (calculated, say, from a budget or standard). To practice our terminology we will rewrite the total variance analysis above as:

$$\text{Total revenue variance} = AR - ER = \$8,200 \text{ (F)}$$

Since we know that revenue, R, can be written as $R = P * Q$, where P is the price per unit, and Q equals the number of units sold, we can write the revenue quantity variance as:

$$\begin{aligned}
 \text{Revenue quantity variance} &= (AQ - EQ) * EP \\
 &= (22,000 - 20,000) * 3.00 \\
 &= 6,000 \text{ (F)}
 \end{aligned}$$

This calculation tells us that revenues should have increased by \$6,000 because of the increase in sales of 2,000 units. But actual revenues increased by \$8,200. The difference is explained by the **price variance** – selling units at an actual price different from that expected:

$$\begin{aligned}
 \text{Revenue price variance} &= (AP - EP) * AQ \\
 &= (3.10 - 3.00) * 22,000 \\
 &= 2,200 \text{ (F)}
 \end{aligned}$$

Steely-eyed readers will notice that the quantity variance is calculated at the expected price, but the price variance is calculated using the actual (not the expected) quantity. This is a convention. Technically, a joint price-quantity variance exists:

$$\text{Joint variance} = (AQ - EQ) * (AP - EP)$$

This joint variance is zero if either the actual quantity equals the expected quantity, or if actual price equals expected price. Normally, neither of these conditions holds. Rather than have endless discussions about whether to assign this joint variance to the quantity or the price variance, accountants have adopted the convention of assigning it to the price variance.

Also, observe that in the numerical example the quantity variance and the price variance sum to the total variance. This is not an accident. It comes from the formula:

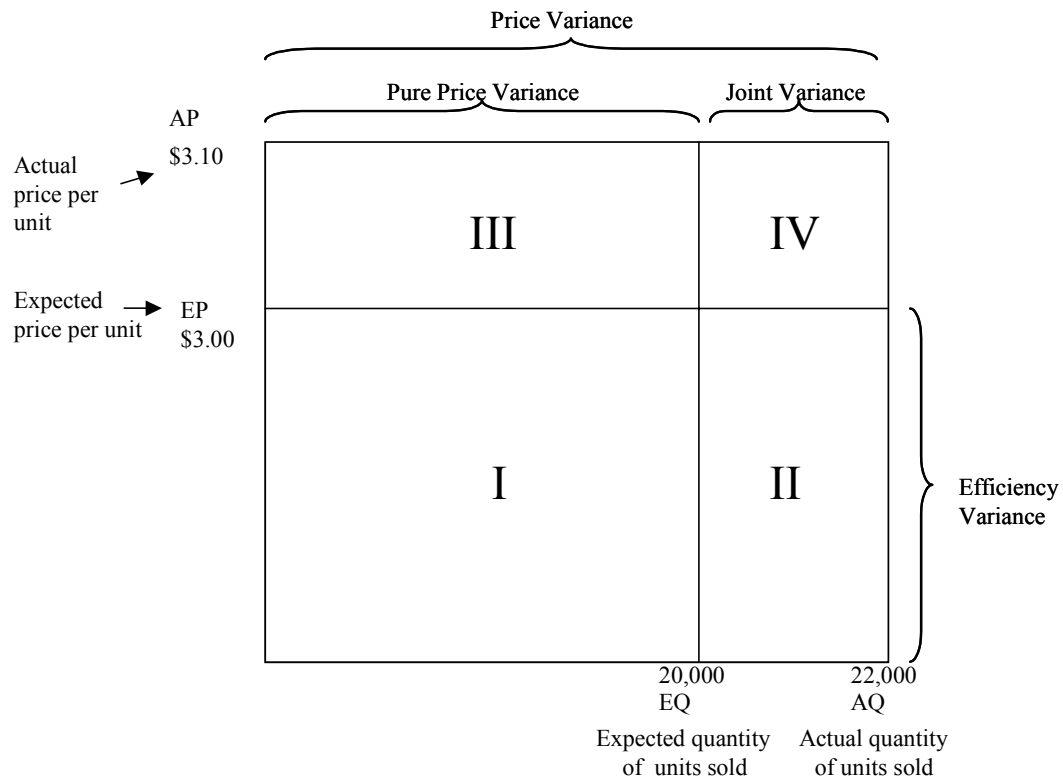
$$\begin{aligned} \text{Total variance} &= AR - ER = (AQ * AP) - (EQ * EP) \\ &= \text{Price variance} + \text{Quantity variance} \\ &= [AQ * (AP - EP)] + [(AQ - EQ) * EP] \end{aligned}$$

Some students find it easier to visualize variance analysis with the aid of a diagram. Exhibit 1 shows the setup for analyzing the revenue variance.

The graphical representation is easiest to interpret when;

- a. the actual price (AP) exceeds the expected price (EP), and
- b. the actual quantity (AQ) exceeds the expected quantity (EQ)

In this case, four non-overlapping areas can be distinguished. When either (a) or (b) does not hold, the graphical representation is more difficult to understand because the four areas overlap. The algebraic formula, however, is always valid.

Exhibit 1 Variance Analysis of Revenues

AREA	FORMULA	DESCRIPTION
I	$EP \times EQ$ $\$3.00 \times 20,000$	Total expected revenue
II	$(AQ - EQ) \times EP$ $22,000 - 20,000 \times \$3.00$	Quantity variance
III	$(AP - EP) \times EQ$ $(\$3.10 - \$3.00) \times 20,000$	Pure price variance
IV	$(AP - EP) \times (AQ - EQ)$ $(\$3.10 - \$3.00) \times (22,000 - 20,000)$	Joint price-quantity variance

Price variance

You can try to draw the four areas in the diagram when selling 22,000 units at an actual price of \$2.90 per unit to yield actual revenues are \$63,800,. The algebraic formulas decompose the total revenue variance of \$3,800 as:

$$\text{Revenue quantity variance} = (22,000 - 20,000) \times \$3.00 = \$6,000 \text{ (F)}$$

$$\text{Revenue price variance} = 22,000 \times (2.90 - 3.00) = 2,200 \text{ (U)}$$

Locating these amounts on the variance analysis diagram, however, is not trivial (Hint: the joint variance becomes part of both the quantity and the price variance).

The decomposition into quantity and price components works for costs as well as revenues. Let's analyze energy costs during the period. Recall that we expected a cost of \$18,000, from using 4,000 kwh at a rate of \$ 4.50/kwh. Suppose actual costs were \$19,550, based on using 4,250 kwh at an actual rate of \$4.60/kwh.

$$\text{Total expense variance} = 19,550 - 18,000$$

$$= 1,550 \text{ (U)}$$

$$\text{Quantity variance} = (AQ - EQ) \times EP$$

$$= (4,250 - 4,000) \times 4.50$$

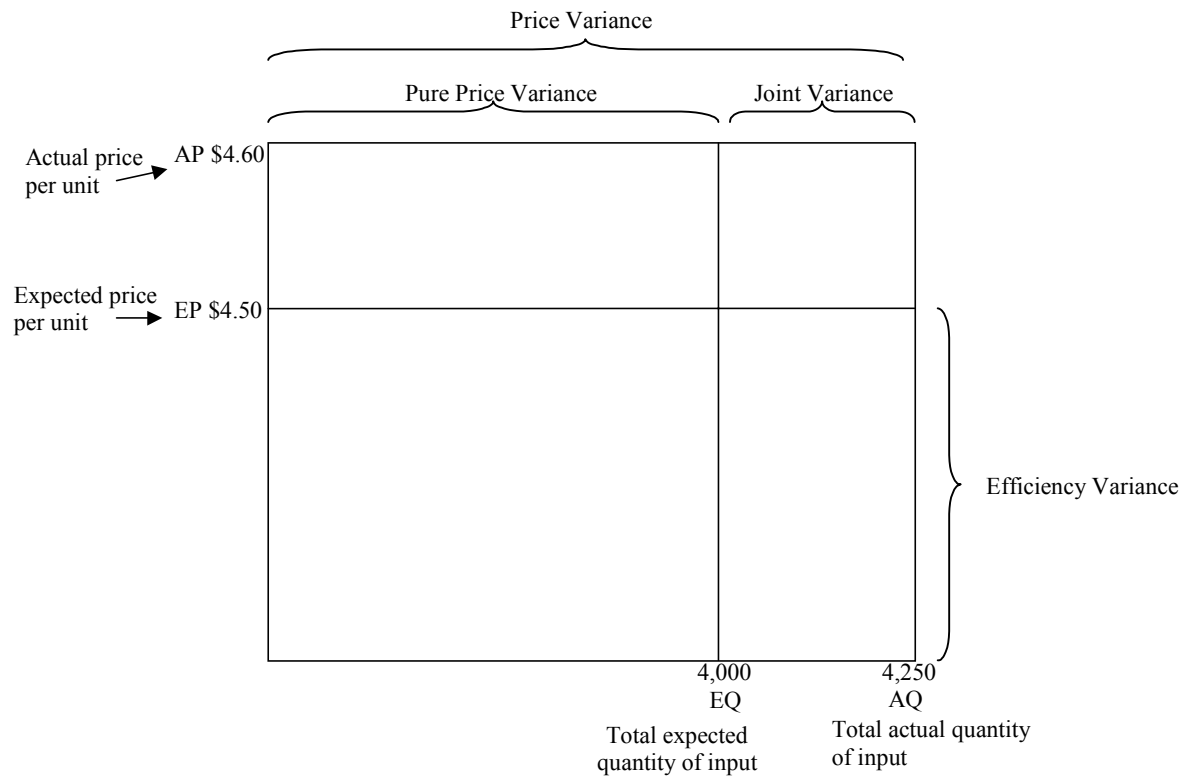
$$= 1,125 \text{ (U)}$$

$$\text{Price variance} = (AP - EP) \times AQ$$

$$= (4.60 - 4.50) \times 4,250$$

$$= 425 \text{ (U)}$$

Graphically, the situation is represented in Exhibit 2:

Exhibit 2 Variance Analysis of Costs

AREA	FORMULA	DESCRIPTION
I	$\text{EP} \times \text{EQ}$ $\$4.50 \times 4,000$	Total budgeted cost
II	$(\text{AQ} - \text{EQ}) \times \text{EP}$ $(4,250 - 4,000) \times \$4.50$	Efficiency variance
III	$(\text{AP} - \text{EP}) \times \text{EQ}$ $(\$4.60 - \$4.50) \times 4,000$	Pure price variance
IV	$(\text{AP} - \text{EP}) \times (\text{AQ} - \text{EQ})$ $(\$4.60 - \$4.50) \times (4,250 - 4,000)$	Joint price-quantity variance

Price variance

When dealing with cost items, the quantity variance is called an **efficiency variance** since it represents the effect of using more or less of an input (materials, labor, or energy) than the standard amount. The price variance is often referred to as a **rate variance**, the effect of using inputs at a higher price than anticipated.

The benefit from separating a total variance into its price (rate) and quantity (efficiency) components occurs because different managers may be responsible for the two variances. The purchasing manager is responsible for acquiring inputs at the standard or budgeted price, and the operations manager is responsible for using the inputs efficiently (not using more than the budgeted quantity). The variance analysis enables us to get a first order look as to whether a materials cost over-run is due more to purchasing behavior than to operating efficiencies. Of course, in real life, assigning responsibility is never quite this simple. A purchasing manager may acquire cheap materials, and earn a favorable purchase price variance. But the operations manager struggles to produce good output from the low-quality materials supplied, and consequently uses more labor and machine time than budgeted, leading to an unfavorable efficiency variance. Who is responsible for the unfavorable efficiency variance – the operations manager or the purchasing manager?

For the revenue variance, the quantity variance might be the responsibility of the sales force, while the marketing group and product manager might have responsibility for setting the price and approving discounts off list price. Economists, however, have taught us that there might be a trade-off between the selling price and the quantity sold. The revenue price and quantity variances help us to determine whether decreases in price are more than compensated by increases in revenues from more units sold.

In summary, quantity and price variances are the building blocks for variance analysis. They illustrate how differences between actual and expected results can be decomposed into price and quantity components, with different questions being asked to different managers as to the source of each variance.

Flexible Budget for Indirect and Support Expenses

Typically, we calculate quantity, price, and mix variances for expenses that can be decomposed into quantity and price components. Examples include the cost of materials, the cost of direct labor, and the cost of other resources, such as energy consumption, where the **quantity** of input is proportional to output. Overhead costs (or as we prefer to call them, indirect and support costs) often don't vary proportionally with output. Also, many of these cost categories – advertising, information services, postage, and telecommunications – are managed at the aggregate spending level for the category, not at the individual resource level. If all these expenses are committed in advance, and not expected to vary with the volume and mix of production or sales during the period, then variance analysis is simple. The static approach of comparing actual to expected expenses would be adequate for cost monitoring and cost control.

Often, however, managers expect that many of their indirect and support expenses should be managed or controlled based on actual activity levels during the period. Thus, some of these expenses are expected to vary with output levels during the period. To deal with indirect expenses that have both fixed (committed) components each period, independent of the volume of activity, and variable (flexible) components that vary with output levels, accountants have created the **flexible budget**. The flexible budget estimates the expected cost at each possible output level. In general, a flexible budget can accommodate jumps in fixed costs once output exceeds the capacity of any

resource as well as pure variable costs, and mixtures of fixed and variable cost. As an example of a simple flexible budget consider the following:

Flexible Budget

	"Normal" ↓					
Capacity Utilization:	40% ^a	60%	70%	80%	100%	
Labor cost (piece rate):	\$4,000	\$6,000	\$7,000	\$8,000	\$10,000	
Allowed Overhead						Cost behavior
Rent	\$500	\$500	\$500	\$500	\$500	Pure nonvariable or "fixed"
Depreciation	500	500	500	500	500	Pure nonvariable or "fixed"
Supervision	500	1,000	1,000	1,000	1,500	"Step" cost ^b
Supplies	400	600	700	800	1,000	Pure variable at 10% of labor
Power	600	700	750	800	900	Semivariable \$400 + 5% of labor
Total	\$2,500	\$3,300	\$3,450	\$3,600	\$4,400	

^aIt is assumed that volume would never fall below 40% of capacity.

^bA "step" cost is one which does not vary directly with production but does increase in lump-sum jumps when volume rises substantially. An example would be adding a second supervisor when volume rose to the level where a second shift was needed.

The simplest form of a flexible budget consists of a fixed component that is independent of production or sales volume, and a variable component that is proportional to some measure of activity volume. In the numerical example above, if we assume that production is always between 60 and 80% of capacity, then supervision is a fixed cost at \$1,000, and we can represent the flexible budget in this region as:

$$\text{Flexible budget} = \$2,400 + 15\% \text{ Direct Labor } \$$$

In this example, the measure of activity volume is Direct Labor \$.

With a flexible budget, we can now analyze difference between actual and budgeted indirect expenses. For example, assume that the company expected to operate at 80% of capacity, where direct labor cost is \$8,000. Its budget for indirect expenses is \$3,600 (either from the table or the formula). Suppose now that the actual direct labor cost for the period was \$7,400 and that actual indirect expenses were \$3,550.

The static budget shows a favorable variance of \$ 50, since actual expenses were below the budgeted amount. But before passing out bonus checks to the manufacturing manager, we should first compute the flexible budget at the **actual** level of activity for the period. The budget is called "flexible" since it varies (or "flexes") with changes from the expected level of activity that was used to draw up the original budget. The flexible budget when DL\$ = \$7,400 is:

$$\begin{aligned}\text{Flexible budget} &= \$2,400 + 15\% (7,400) \\ &= \$3,510.\end{aligned}$$

Relative to this flexible budget, the company actually had an unfavorable **spending** variance:

$$\begin{aligned}\text{Spending variance} &= \text{Actual indirect expenses} - \text{Flexible budget @ actual volume} \\ &= \$3,550 - 3,510 \\ &= \$40 \text{ U}\end{aligned}$$

But we also need to account for the change in expected costs due to working a lower number of direct labor \$. This is called a volume variance. The **volume variance** calculates the difference in costs caused by working fewer direct labor \$ than anticipated. (In general, the volume variance would be calculated based on whatever volume measure is used to authorize or allocate overhead costs.):

$$\begin{aligned}\text{Volume variance} &= (AQ - EQ) * \text{expected variable cost per unit} \\ &= (7,400 - 8,000) * 15\% = \$90 \text{ (F)}\end{aligned}$$

We can now represent the total indirect expense variance of \$ 50 (F) as the sum of a favorable volume variance, working fewer hours than expected and thereby lowering expected costs, and the unfavorable spending variance (relative to the hours actually worked):

$$\begin{aligned}\text{Total indirect expense variance} &= \$50 \text{ (F)} \\ &= \text{Volume variance } (\$90 \text{ F}) + \text{Spending variance } (-\$40 \text{ U})\end{aligned}$$

The volume variance is generally considered uncontrollable for operations managers, but these managers are held accountable for the spending variance, calculated from the flexible budget. The flexible budget identifies those expenses that the manager is expected to reduce when actual activity volumes are decreasing, and, conversely, the expenses that can increase beyond the static budget when actual production or sales exceeds the budgeted quantity.

Mix Variance

In the above example, we analyzed a situation with only a single product. Suppose, in a more realistic situation, the company sells multiple products and services. Then we have an additional complication. Revenues may differ from expectations even if there is no variation in the total number of units sold or in the individual selling prices. This situation arises when the mix of product shifts between high and low priced items. Consider first an aggregate revenue statement.

	<u>Budgeted</u>	<u>Actual</u>
Average Selling Price	\$ 4.50	\$4.20
Quantity	6,500	6,850
Total Revenues	\$ 29,250	\$28,770

We start by calculating our usual quantity and price variances.

$$\begin{aligned}\text{Revenue quantity variance} &= (AQ - EQ) * EP \\ &= (6,850 - 6,500) * \$4.50 \\ &= \$1,575 \text{ (F)}\end{aligned}$$

$$\begin{aligned}
 \text{Revenue price variance} &= AQ * (AP - EP) \\
 &= 6,850 * (\$4.20 - \$4.50) \\
 &= (\$2,055) \text{ (U)}
 \end{aligned}$$

Thus the unfavorable revenue variance of \$480 is the sum of a favorable quantity variance of \$1,575 (from selling more units than budgeted) and an unfavorable price variance of \$2,055.

Now, let's go behind the aggregate numbers. We learn that the company sells three different products, with the budgeted revenues shown below:

Expected Sales Product	X1	X2	X3	Total
Selling Price	\$ 3.00	\$ 4.00	\$ 7.50	\$ 4.50
Selling Quantity	2,000	3,000	1,500	6,500
Revenues	\$ 6,000	\$ 12,000	\$ 11,250	\$ 29,250

The actual total sales of 6,850 units, more than 5% higher than budgeted, were distributed across the three products as shown below:

Actual Sales Product	X1	X2	X3	Total
Selling Price	\$ 3.1607	\$ 4.10	\$ 8.00	\$ 4.20
Selling Quantity	2,800	3,200	850	6,850
Revenues	\$ 8,850	\$ 13,120	\$ 6,800	\$ 28,770

We now have a puzzle. Despite selling 5% more units than budgeted, and with each product selling at higher prices than budgeted, total revenues have decreased. Also, how, with the actual prices of each product exceeding the budgeted prices, can the company show an unfavorable price variance, as we calculated above?

Intuitively, total revenues declined because sales increased in the low-priced products, X1 and X2, while sales of the high-priced product, X3, plummeted by nearly 45%. Calculating a mix variance helps to quantify this effect. It can be applied even in situations where there are hundreds or thousands of different products and services sold.

We start by calculating a price variance for the sales of the individual products:

$$\begin{aligned}
 \text{Price variance} &= AQ * (AP - EP) \\
 &= 2,800 * (3.1607 - 3.00) + 3,200 * (4.10 - 4.00) + 850 * (8.00 - 7.50) \\
 &= \$ 1,195 \text{ (F)}
 \end{aligned}$$

The favorable price variance of \$1,195 is plausible since the prices of all units increased.

To examine the impact of mix shifts, we require a new variable to be calculated, the expected average selling price (EASP). From the table of expected sales, we calculate EASP by dividing total expected revenues by total expected units sold:

$$\begin{aligned}
 \text{Average Selling Price (EASP)} &= (\text{Total revenues}) / (\text{Total units sold}) \\
 &= \$29,250 / 6,500 \\
 &= \$4.50
 \end{aligned}$$

(Of course this \$4.50 average price must be the same as the aggregate price we used in our initial calculations.)

We calculate the mix variance by examining the shift in sales for each product, and multiplying by the difference between the product's expected price and the average selling price:

$$\begin{aligned}
 \text{Mix variance} &= \sum_i (AQ_i - EQ_i) * (EP_i - EASP_i) \\
 &= (2,800 - 2,000) * (3.00 - 4.50) + \\
 &\quad (3,200 - 3,000) * (4.00 - 4.50) + \\
 &\quad (850 - 1,500) * (7.50 - 4.50) \\
 &= (-1,200) + (-100) + (-1,950) \\
 &= 3,250 \text{ (U)}.
 \end{aligned}$$

The highly unfavorable mix variance of \$3,250 is the sum of a negative mix variance for each product. It is unfavorable because the company sold:

- more of the products whose prices were lower than EASP, the average selling price, and
- fewer of the products whose prices exceeded EASP.

Conversely, the mix variance will be favorable when the company sells more of the high-priced items and less of the low-priced items (measured relative to the average selling price).

In our example, if the revenue mix had stayed constant, we would have enjoyed a favorable price variance of \$1,195 from higher prices for each product. But the unfavorable shift in the mix of products sold cost us \$3,250. The net price variance of \$2,055 (U) is the algebraic sum of these two.